CENG 301 – Database Systems Project Report

Chosen Topic: Gym Management System

1. Introduction

This project was developed for the CENG 301 Database Systems course. The main goal of the project is to design and implement a Gym Management System that uses a relational database and a simple web-based user interface. The system allows administrators to manage members, trainers, classes, memberships, equipment, payments, and reports.

The project focuses on:

- Proper database design (tables, relationships, constraints)
- Use of SQL queries and stored procedures
- Connection between a backend application and a SQL Server database
- A graphical user interface (GUI) instead of a console application

The application is implemented using C# (.NET Minimal API) and Microsoft SQL Server.

2. System Overview

The system is designed as a web-based application. Users interact with the system through HTML pages served by the backend. All data operations are handled through SQL queries and stored procedures.

Main Features

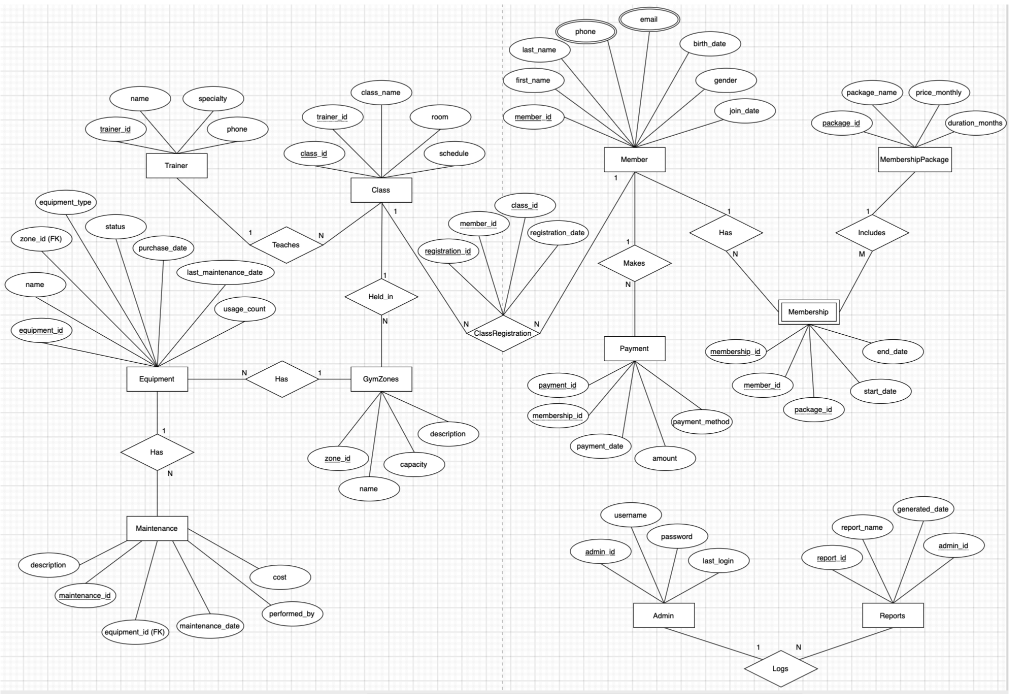
- Admin login system
- Member management (add, list, delete)
- Trainer and class management
- Membership packages and memberships
- Equipment and maintenance tracking
- Payment records
- Reports generated using stored procedures

3. Technologies Used

- **Backend:** C# (.NET Minimal API)
- **Database:** Microsoft SQL Server
- **Frontend:** HTML & CSS
- **Data Access**: ADO.NET (SqlConnection, SqlCommand)

This technology stack was chosen because it is simple, efficient, and suitable for a database-focused course project.

4. Entity Relationship (ER) Diagram



This figure shows the Entity Relationship (ER) diagram of the Gym Management System. The diagram includes the main entities, their attributes, and the relationships between them.

5. ER to Relational Mapping

In this stage, the ER diagram of the Gym Management System is transformed into a relational database schema. Each entity in the ER model is mapped to a table, and relationships are represented using primary and foreign keys.

- Each entity such as Admin, Member, Trainer, Class, Membership, Payment, Equipment, Maintenance, and Gym_Zone is mapped to a relational table with its own primary key.

- One-to-many (1:N) relationships are implemented by placing the primary key of the parent entity as a foreign key in the related table. For example, the relationship between Trainer and Class is represented using trainer_id as a foreign key in the Class table.

- The many-to-many (M:N) relationship between Member and Class is resolved using the Class_Registration table, which contains foreign keys referencing both entities.

- Referential integrity is enforced using foreign key constraints to maintain data consistency.

This mapping ensures that the conceptual ER model is correctly transformed into a relational schema suitable for implementation in SQL.

6. Table / Schema Descriptions

This section describes the database tables of the Gym Management System, including their primary keys (PK) and foreign key (FK) relationships.

- **Admin (admin_id, PK):** Stores administrator information. Referenced by the Reports table.
- **Reports (report_id, PK, admin_id, FK)**: Stores reports generated by administrators.
- **Member (member_id, PK):** Stores gym member information. Referenced by Class_Registration and Membership tables.
- **Trainer (trainer_id, PK):** Stores trainer details. Referenced by the Class table.
- **Class (class_id, PK, trainer_id, FK):** Stores gym class information and assigned trainers.
- **Class_Registration (registration_id, PK, member_id, FK, class_id, FK):** Resolves the many-to-many relationship between members and classes.
- **Membership_Package (package_id, PK):** Stores membership package information.
- **Membership (membership_id, PK, member_id, FK, package_id, FK):** Stores membership details of members.
- **Payment (payment_id, PK, membership_id, FK):** Stores payment records for memberships.
- **Gym_Zone (zone_id, PK)**: Stores different zones of the gym.
- **Equipment (equipment_id, PK, zone_id, FK):** Stores gym equipment information.
- **Maintenance (maintenance_id, PK, equipment_id, FK):** Stores maintenance records for equipment.

Foreign key constraints are used to maintain relationships between tables, and cascading rules are applied where necessary to ensure data integrity.

7. Database Design

The database is named gym_management_web. It follows a relational model and uses primary keys, foreign keys, constraints, and indexes to ensure data integrity.

8. Backend Implementation

The backend is implemented using Minimal API structure in C#. Each functionality is mapped to a specific HTTP endpoint.

Examples:

- GET /members – Lists all members
- POST /members/add – Adds a new member
- GET /classes – Lists all classes
- POST /create/class – Creates a new class

SQL queries are executed using parameterized commands to prevent SQL injection.

9. Stored Procedures

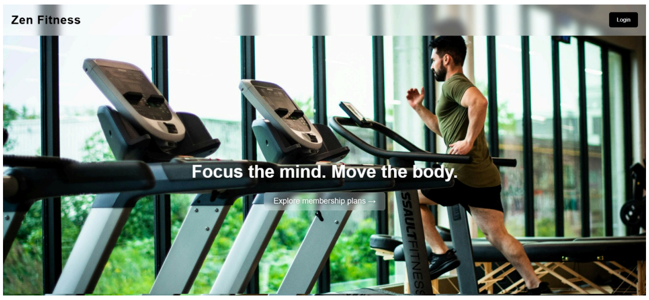
Stored procedures are used for reporting and complex operations. Using stored procedures improves performance and keeps business logic inside the database.

Examples:

- getMostActiveMember: Finds the member who attended the most classes
- getEquipmentUsage: Lists equipment with the highest usage count
- markEquipmentMaintenance: Inserts a maintenance record and updates equipment status

At least one stored procedure is actively used by the backend application during runtime.

10. User Interface (GUI)



The interface is web-based and built using HTML and CSS.

Each page displays:

- Data tables
- Forms for adding or removing records
- Navigation links between pages

This interface satisfies the course requirement for a graphical user interface and ensures that the application is not a console-based system.

← Back to Reports

Most Active Member

Name	Ece Bayyar
Total Classes	4

Class Registrations

ID	Member	Class	Registration Date
1	Burce Nur Kavak	Morning Yoga	2025-12-01
2	Ece Bayyar	Flow & Glow	2025-12-02
3	Ibrahim Said Akinci	CoreForce	2025-12-03
10	Selin Gul Bayri	Morning Yoga	2025-12-20
11	Joshua Hong	Zen Pilates	2025-12-15
12	Mert Demir	Lune Dance	2025-12-09
13	Burce Nur Kavak	Morning Yoga	2025-12-10
14	Burce Nur Kavak	Flow & Glow	2025-12-11
15	Burce Nur Kavak	CoreForce	2025-12-12
16	Ece Bayyar	Morning Yoga	2025-12-10
17	Ece Bayyar	Morning Yoga	2025-12-11
18	Ece Bayyar	Zen Pilates	2025-12-12
19	Joshua Hong	Morning Yoga	2025-12-14

Remove Registration

Burce Nur Kavak - Morning Yoga Remove

11. Sample Data

Sample data is inserted into the database to demonstrate system functionality.
The data includes:

- Members
- Trainers
- Classes
- Equipment
- Payments

Insert operations are protected using IF NOT EXISTS statements to avoid duplicate records.

12. Conclusion

In this project, a complete Gym Management System was developed using relational database concepts.
The project demonstrates:

- Correct database modeling
- Use of SQL constraints and indexes
- Integration between C# backend and SQL Server
- Usage of stored procedures
- A functional graphical user interface

The system meets all course requirements and provides a solid example of a real-world database application.

13. Group Work Contribution

This project was developed by a group of four students as part of the CENG 301 Database Systems course. The responsibilities were shared among group members, including database design, backend development, user interface implementation, and testing.

Team Members and Student IDs

- 220446032 - Selin Gül Bayrı
- 230446021 - Ece Bayyar
- 230446041 - Burçe Nur Kavak
- 230446402 - İbrahim Said Akıncı